BCS quantum simulation: paper reproduction

Notebook reprodutível para acompanhar o manuscrito `papers/bcs_quantum_simulation.md`.

Ele assume que os experimentos já foram executados e que as tabelas/figuras existem em `results/`. A última seção traz os comandos para regenerar tudo.

In [1]:
```
from pathlib import Path
import pandas as pd
import matplotlib.pyplot as plt
from IPython.display import Image, display, Markdown

cwd = Path.cwd().resolve()
ROOT = cwd if (cwd / 'results').exists() else cwd.parent
TABLES = ROOT / 'results' / 'tables'
FIGURES = ROOT / 'results' / 'figures'

pd.set_option('display.max_columns', 80)
pd.set_option('display.precision', 6)
```

Parâmetros principais

O caso padrão usado no paper é o Hamiltoniano BCS reduzido com quatro modos de pares.

In [2]:
```
params = {
    'xi': [-1.0, -0.35, 0.35, 1.0],
    'g': 0.8,
    'occupied': [0, 1],
    'time': 2.5,
    'final_time_observables': 6.0,
    'trotter_steps': 8,
    'trotter_order': 2,
}
params
```

Out[2]:
```
{'xi': [-1.0, -0.35, 0.35, 1.0],
 'g': 0.8,
 'occupied': [0, 1],
 'time': 2.5,
 'final_time_observables': 6.0,
 'trotter_steps': 8,
 'trotter_order': 2}
```

Funções auxiliares

In [3]:
```
def read_table(name):
    path = TABLES / name
    if not path.exists():
        raise FileNotFoundError(f'Missing table: {path}')
    return pd.read_csv(path)

def show_figure(name, width=850):
    path = FIGURES / name
    if not path.exists():
        raise FileNotFoundError(f'Missing figure: {path}')
    display(Image(filename=str(path), width=width))

def section(title):
    display(Markdown(f'## {title}'))
```

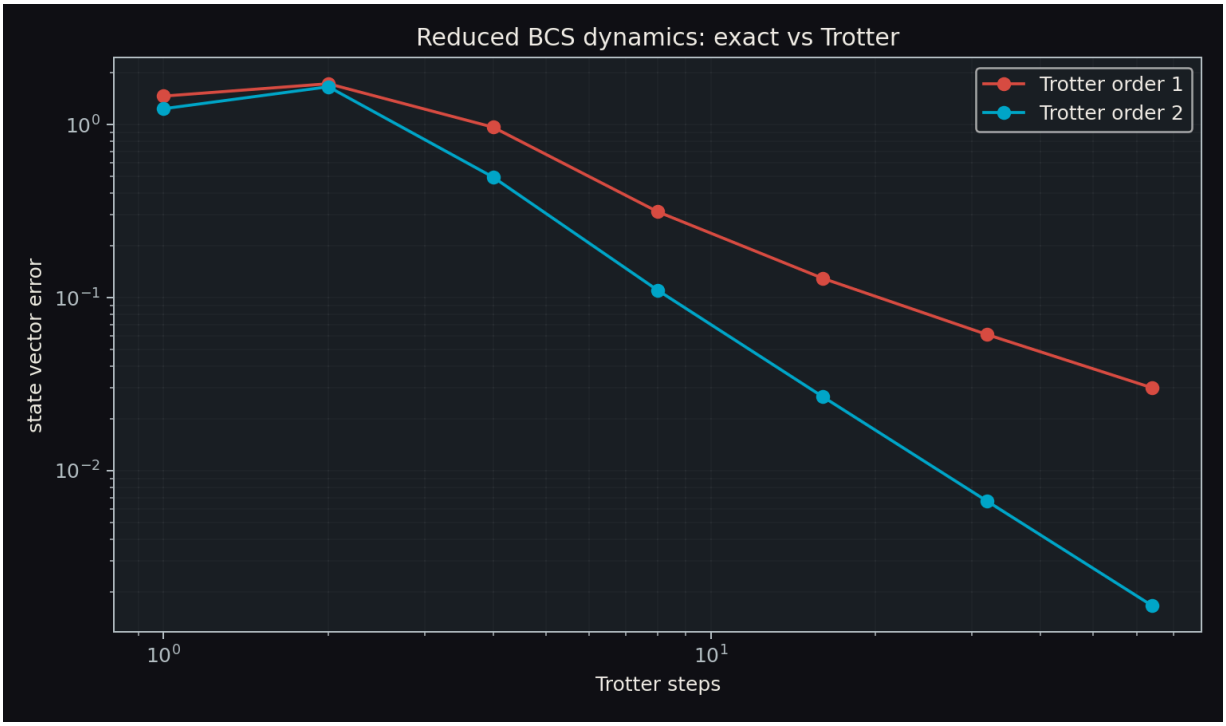
Dinâmica BCS e erro de Trotter

In [4]:
```
trotter = read_table('bcs_dynamics_trotter.csv')
display(trotter)
best = trotter.sort_values('fidelity', ascending=False).iloc[0]
best[['time', 'steps', 'order', 'fidelity', 'state_error', 'exact_gap_proxy'
```

	time	steps	order	fidelity	state_error	exact_energy	trotter_energy	exact_pair_r
0	2.5	1.0	1.0	0.016402	1.459181	2.7	-2.723483	
1	2.5	2.0	1.0	0.422325	1.720592	2.7	0.591860	
2	2.5	4.0	1.0	0.331926	0.964730	2.7	1.138241	
3	2.5	8.0	1.0	0.905103	0.313054	2.7	2.725392	
4	2.5	16.0	1.0	0.983735	0.129330	2.7	2.758153	
5	2.5	32.0	1.0	0.996618	0.061080	2.7	2.736274	
6	2.5	64.0	1.0	0.999226	0.030118	2.7	2.719874	
7	2.5	1.0	2.0	0.684071	1.233649	2.7	2.154363	
8	2.5	2.0	2.0	0.144306	1.652012	2.7	-0.881944	
9	2.5	4.0	2.0	0.858119	0.496573	2.7	1.926304	
10	2.5	8.0	2.0	0.994675	0.110136	2.7	2.611660	
11	2.5	16.0	2.0	0.999706	0.026798	2.7	2.681008	
12	2.5	32.0	2.0	0.999982	0.006655	2.7	2.695413	
13	2.5	64.0	2.0	0.999999	0.001661	2.7	2.698863	

Out[4]:
```
time                2.500000
steps              64.000000
order               2.000000
fidelity            0.999999
state_error         0.001661
exact_gap_proxy     0.803355
trotter_gap_proxy   0.803429
Name: 13, dtype: float64
```

In [5]:
```
show_figure('bcs_trotter_error.png')
```

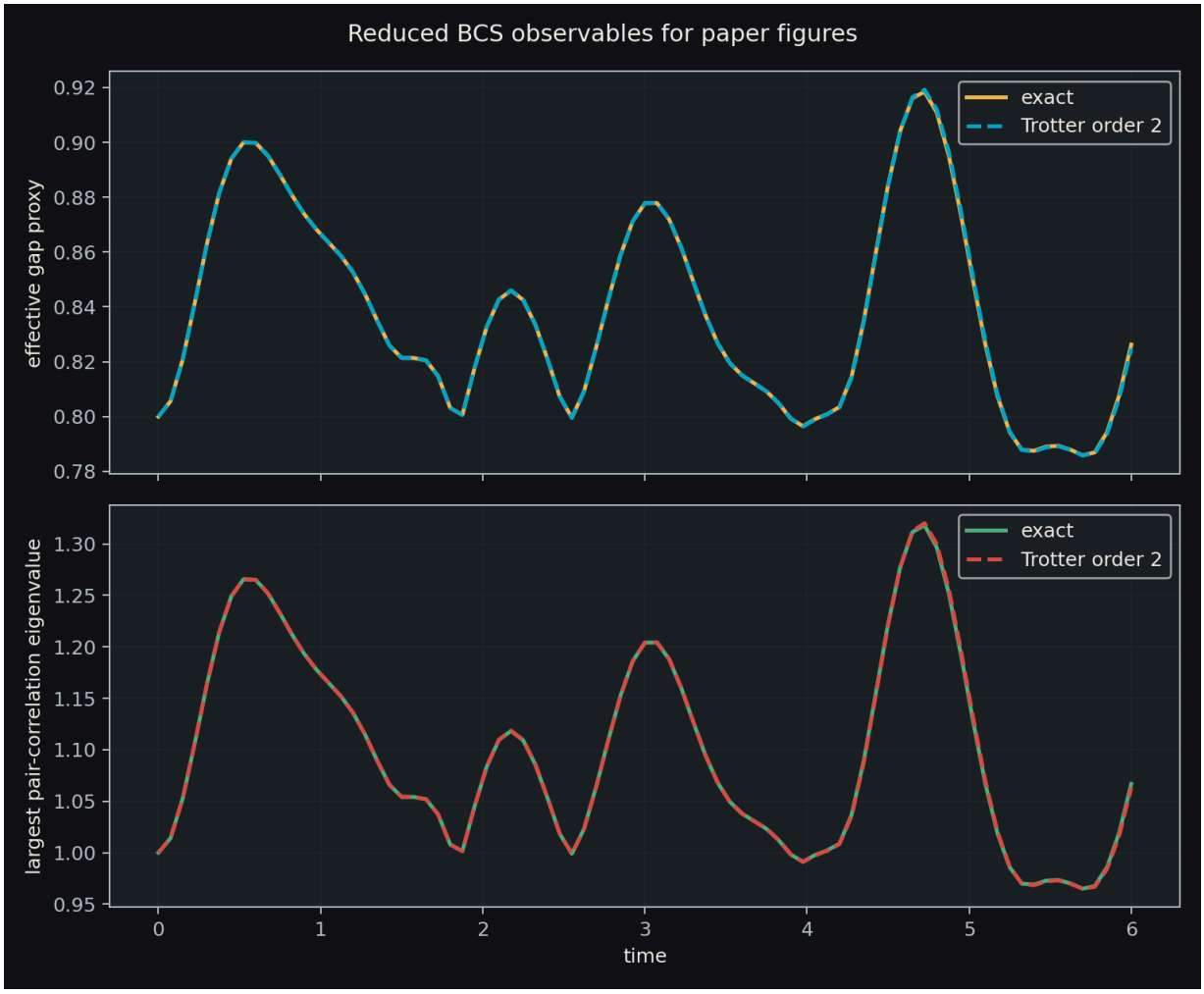


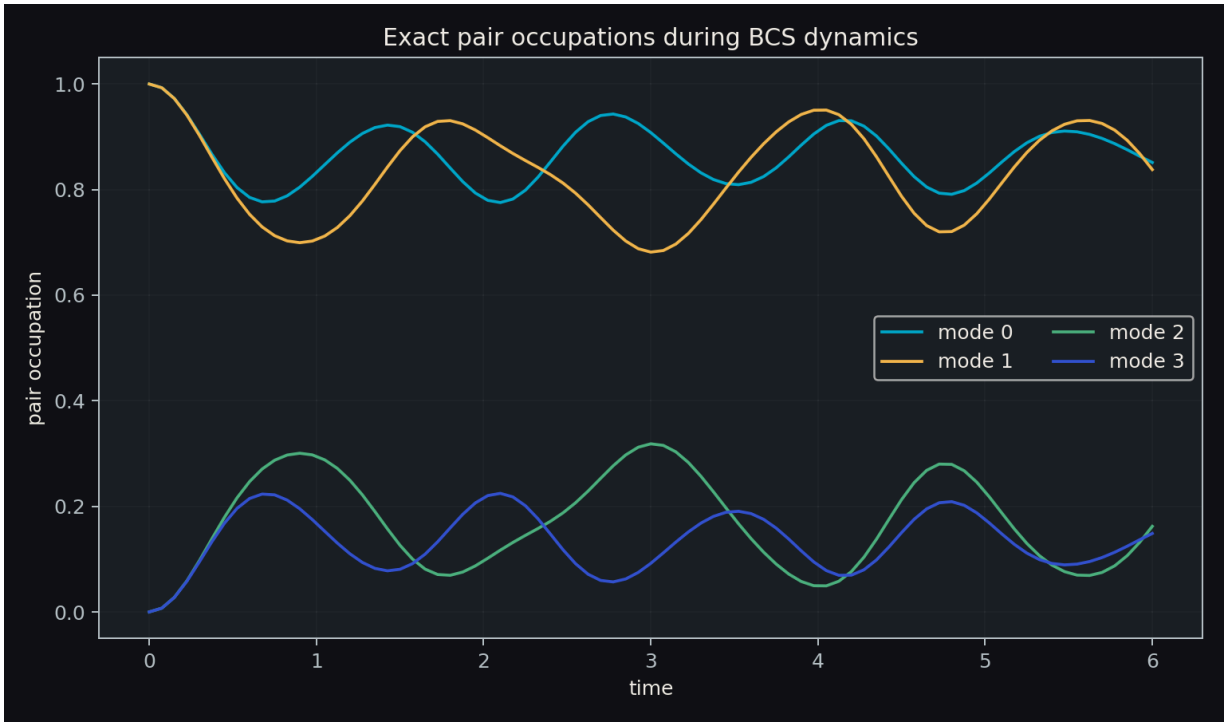
Observáveis físicos para o paper

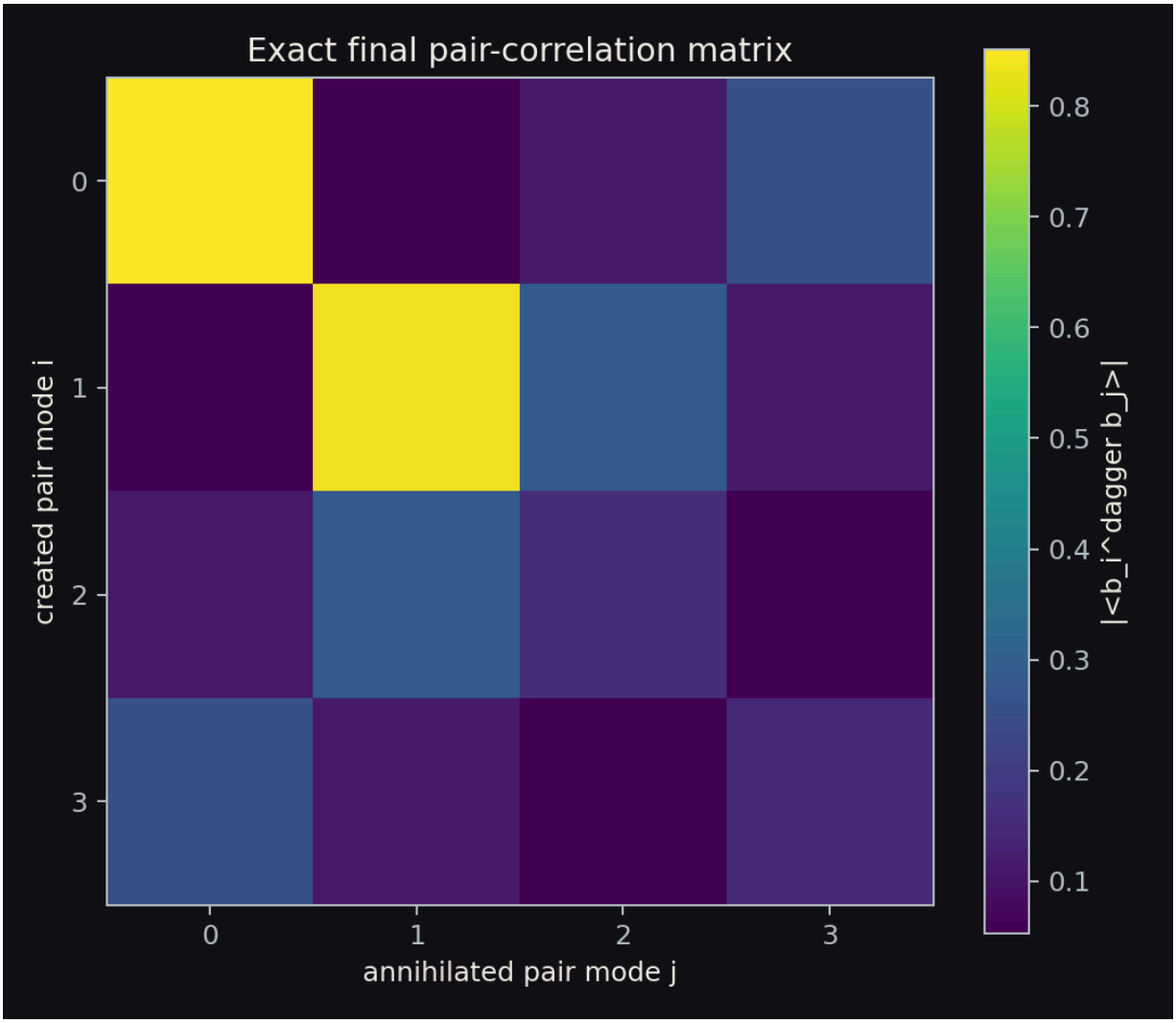
In [6]:
```
obs = read_table('bcs_paper_observables.csv')
final_obs = obs.iloc[-1]
final_obs[['time', 'fidelity', 'state_error', 'exact_pair_number', 'trotter_
```

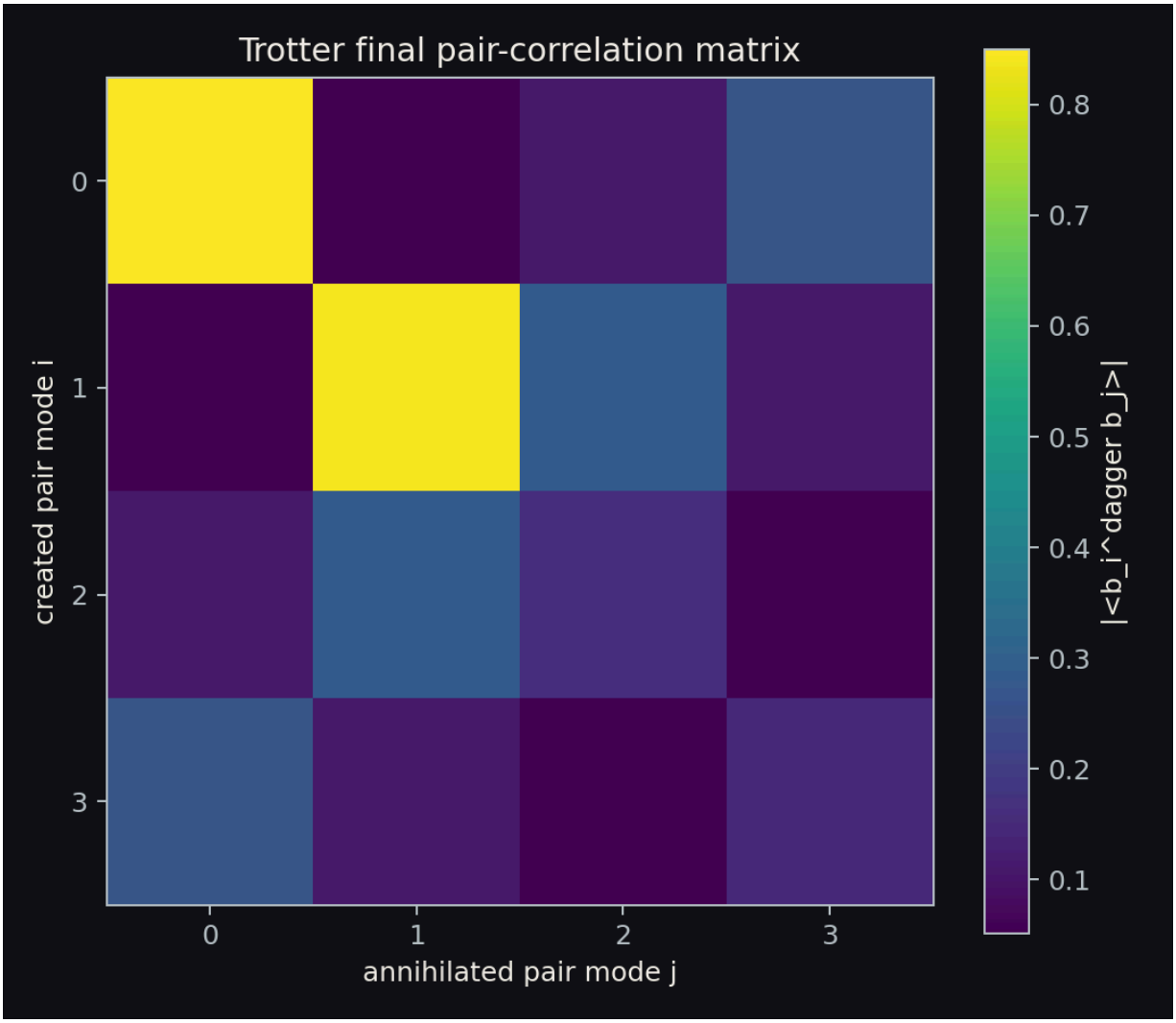
Out[6]:
```
time                 6.000000
fidelity             0.999886
state_error          0.022627
exact_pair_number    2.000000
trotter_pair_number  2.000000
exact_gap_proxy      0.826271
trotter_gap_proxy    0.824833
Name: 80, dtype: float64
```

In [7]:
```
for fig in [
    'bcs_gap_condensate_time.png',
    'bcs_pair_occupations_time.png',
    'bcs_pair_correlation_exact_final.png',
    'bcs_pair_correlation_trotter_final.png',
]:
    show_figure(fig, width=760)
```


Reduced BCS observables for paper figures
effective gap proxy
exact
Trotter order 2
largest pair-correlation eigenvalue
exact
Trotter order 2
time


Exact pair occupations during BCS dynamics
pair occupation
mode 0
mode 1
mode 2
mode 3
time


Exact final pair-correlation matrix
created pair mode i
annihilated pair mode j
|<b_i^dagger b_j>|
0
1
2
3
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1



Green normal, Gor'kov anômalo e matriz Nambu

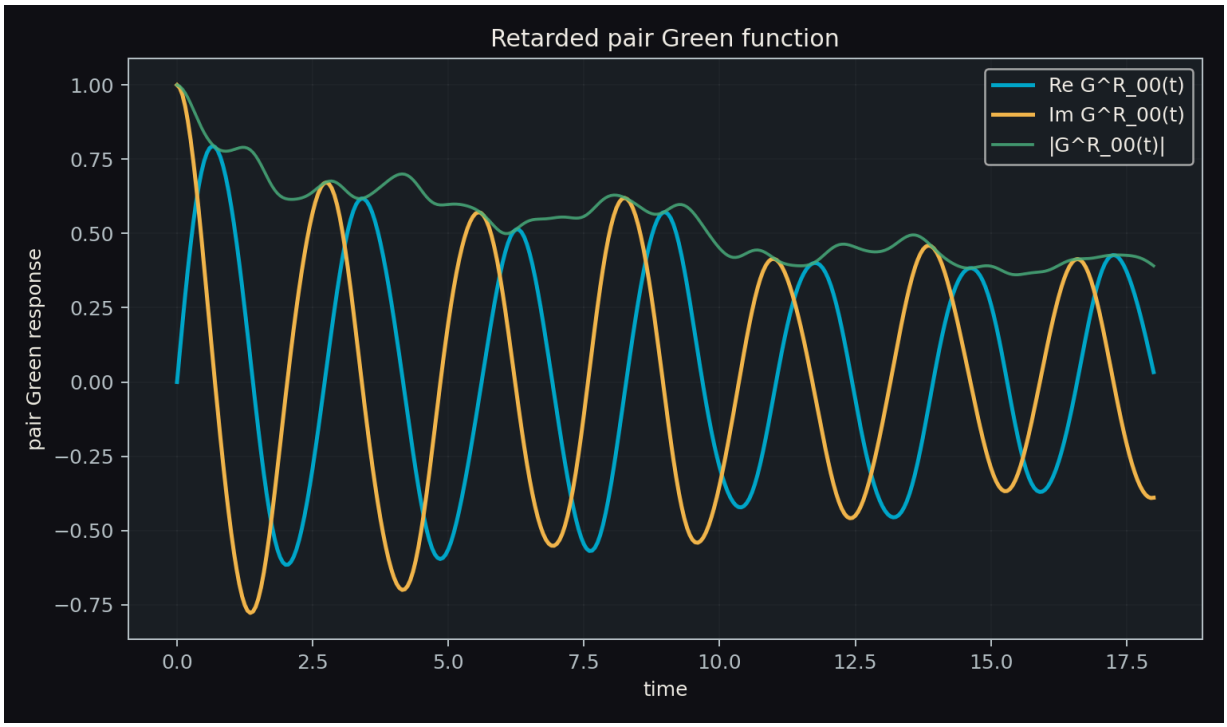
In [8]:
```
green_spec = read_table('bcs_pair_green_spectral_density.csv')
gorkov_spec = read_table('bcs_gorkov_anomalous_spectrum.csv')
nambu_spec = read_table('bcs_gorkov_nambu_spectral_trace.csv')

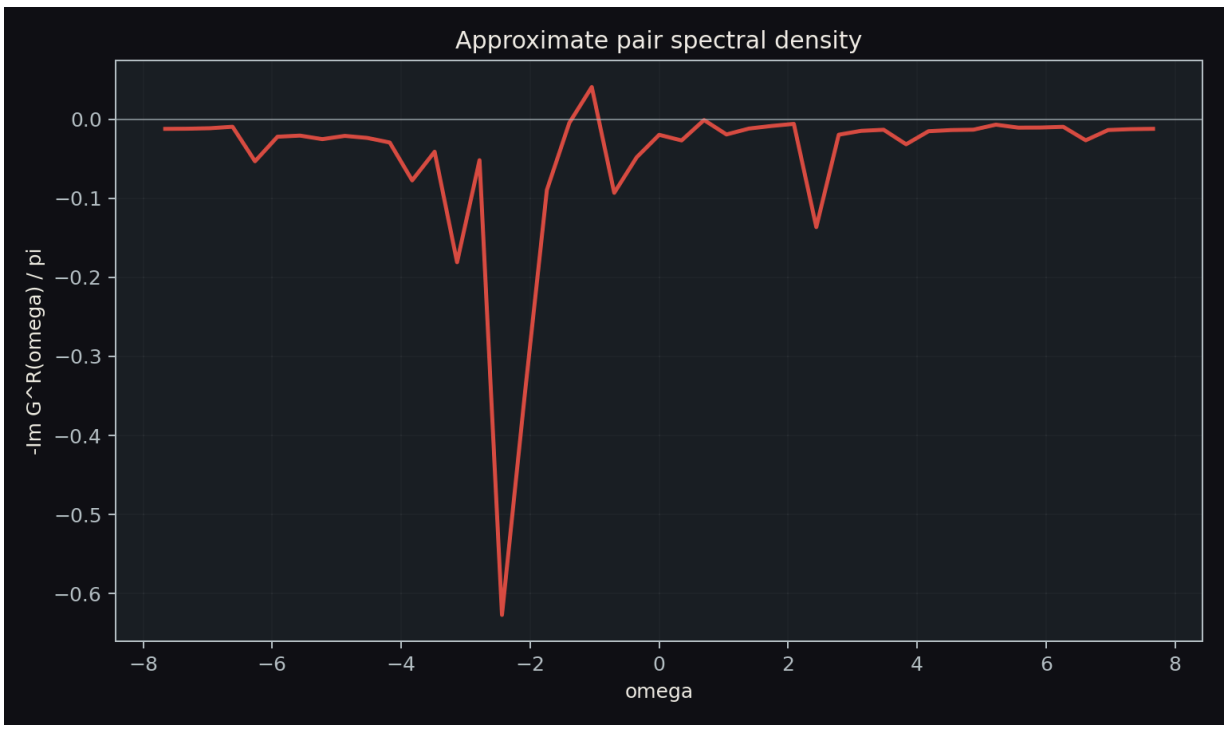
summary_spectral = pd.DataFrame([
    {'channel': 'normal G', **green_spec.sort_values('spectral_weight', asce
    {'channel': 'anomalous F', **gorkov_spec.sort_values('spectral_weight',
    {'channel': 'Nambu trace', **nambu_spec.sort_values('spectral_weight', a
])
summary_spectral
```

Out[8]:

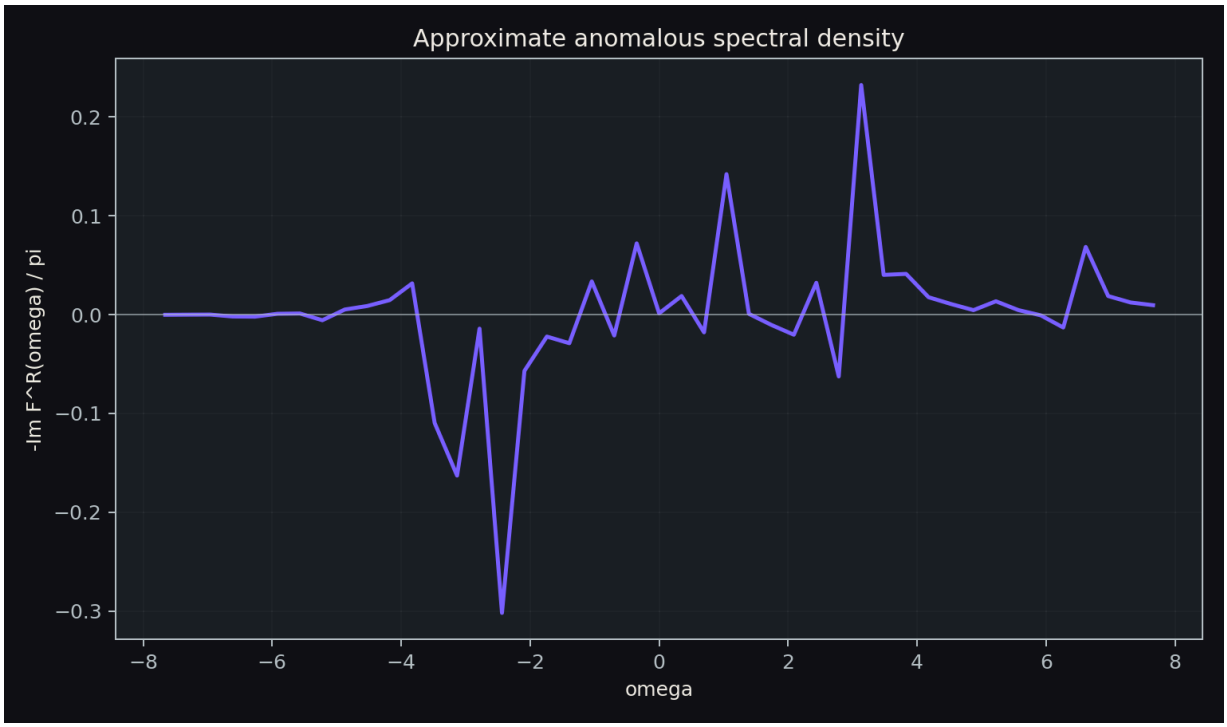
	channel	omega	spectral_weight
0	normal G	-1.044297	0.040784
1	anomalous F	3.132890	0.232452
2	Nambu trace	-2.436692	0.673764

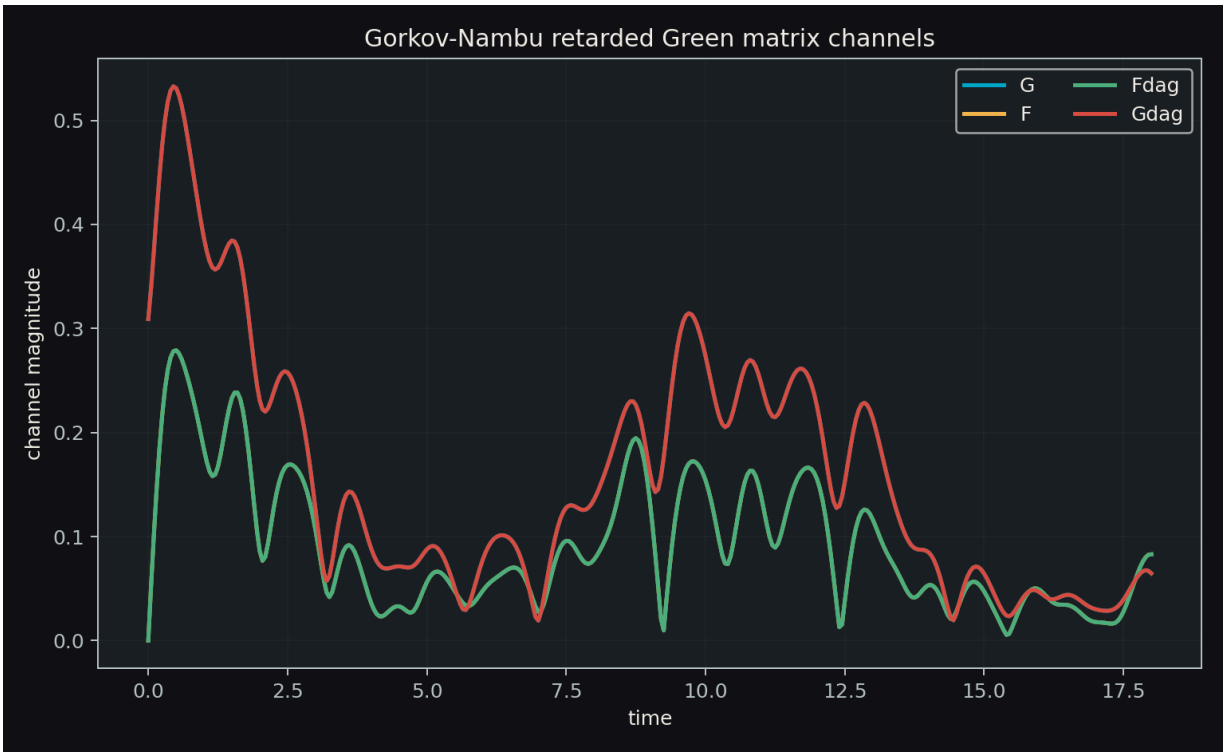
In [9]:
```
for fig in [
    'bcs_pair_green_time.png',
    'bcs_pair_green_spectrum.png',
    'bcs_gorkov_anomalous_time.png',
    'bcs_gorkov_anomalous_spectrum.png',
    'bcs_gorkov_nambu_channels.png',
    'bcs_gorkov_nambu_spectral_trace.png',
]:
    show_figure(fig, width=820)
```

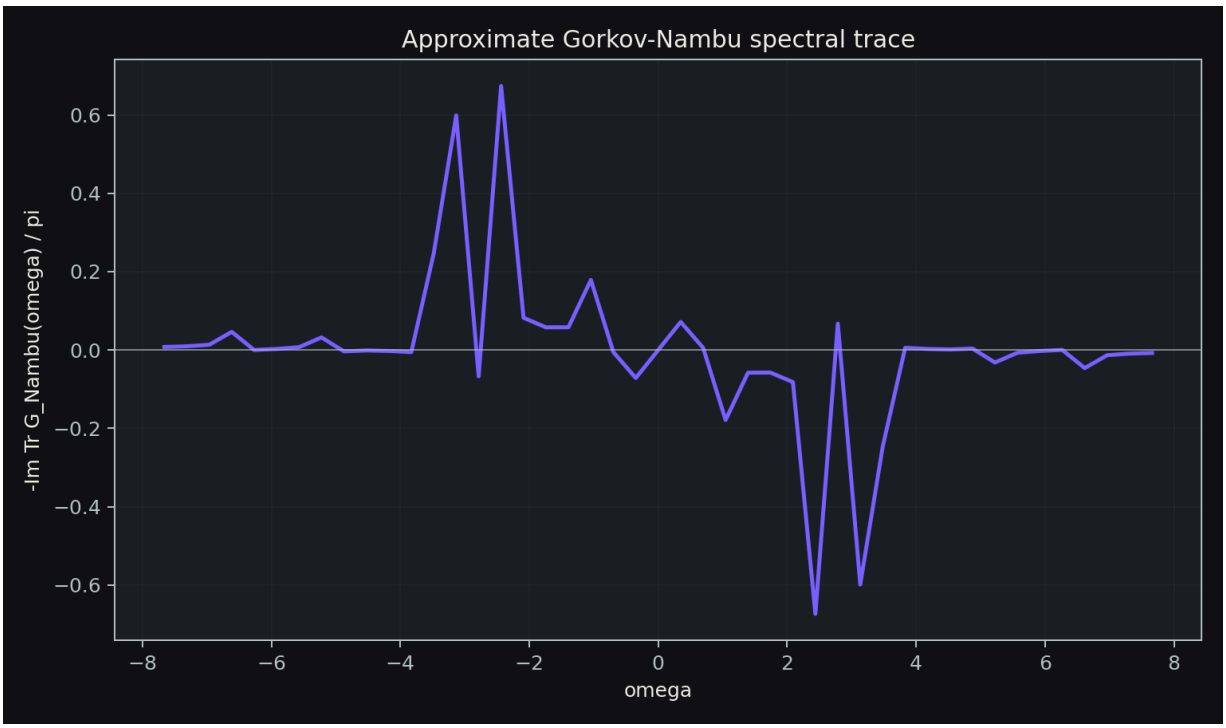





Gorkov anomalous retarded Green function
Re F^R_00(t)
Im F^R_00(t)
|F^R_00(t)|
anomalous pair response
time
0.3
0.2
0.1
0.0
−0.1
−0.2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Approximate anomalous spectral density
-Im F^R(omega) / pi
omega
0.2
0.1
0.0
−0.1
−0.2
−0.3
−8
−6
−4
−2
0
2
4
6
8





Qiskit shots, energia por Pauli e mitigação

In [10]:
```
shots = read_table('bcs_qiskit_shots_noise.csv')
energy = read_table('bcs_qiskit_energy_shots.csv')
mitigation = read_table('bcs_qiskit_error_mitigation.csv')

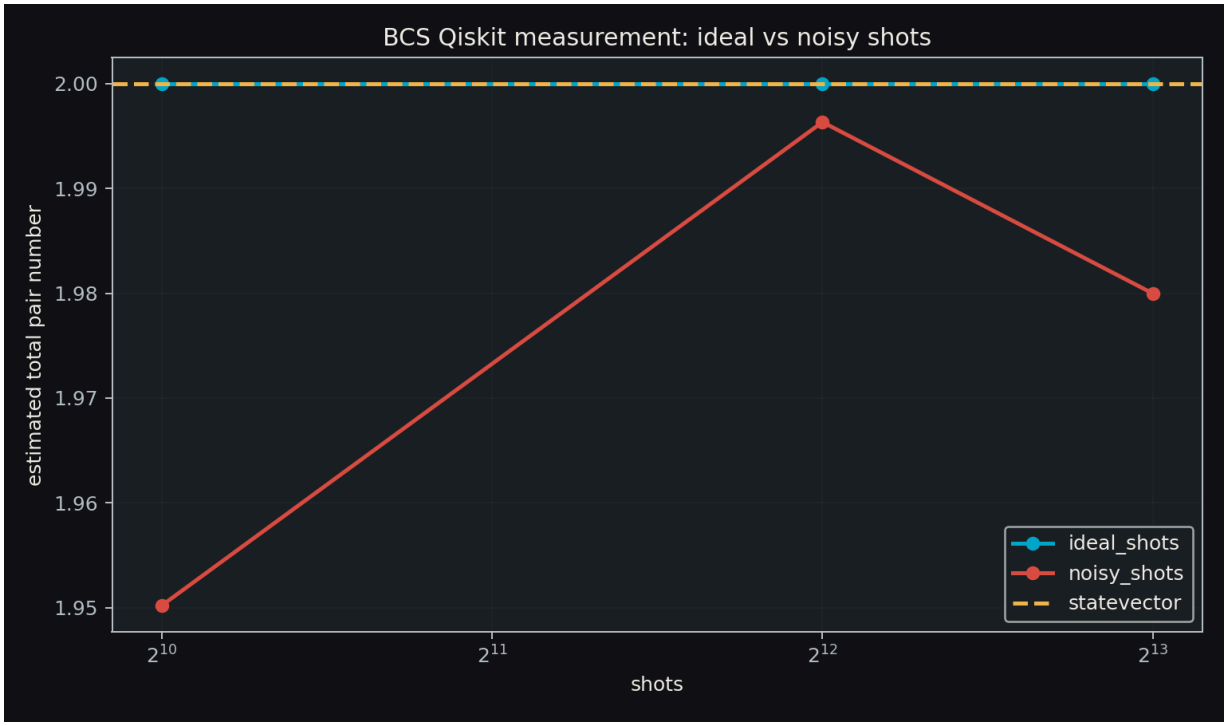
display(shots[['mode', 'shots', 'total_pair_number', 'total_pair_error']])
display(energy[['mode', 'shots', 'exact_energy', 'statevector_energy', 'ener
display(mitigation[['mode', 'noise_scale', 'energy', 'abs_error_vs_statevect
```

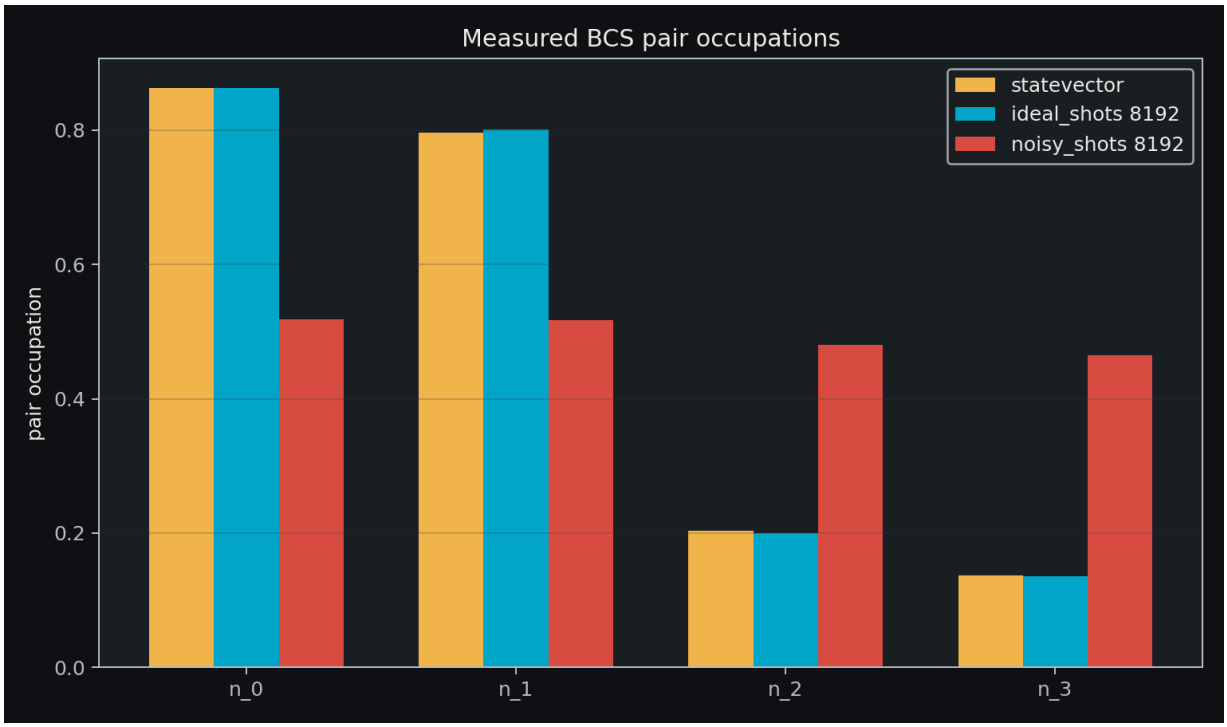
	mode	shots	total_pair_number	total_pair_error
0	ideal_shots	1024.0	2.000000	3.330669e-14
1	noisy_shots	1024.0	1.950195	4.980469e-02
2	ideal_shots	4096.0	2.000000	3.330669e-14
3	noisy_shots	4096.0	1.996338	3.662109e-03
4	ideal_shots	8192.0	2.000000	3.330669e-14
5	noisy_shots	8192.0	1.979980	2.001953e-02

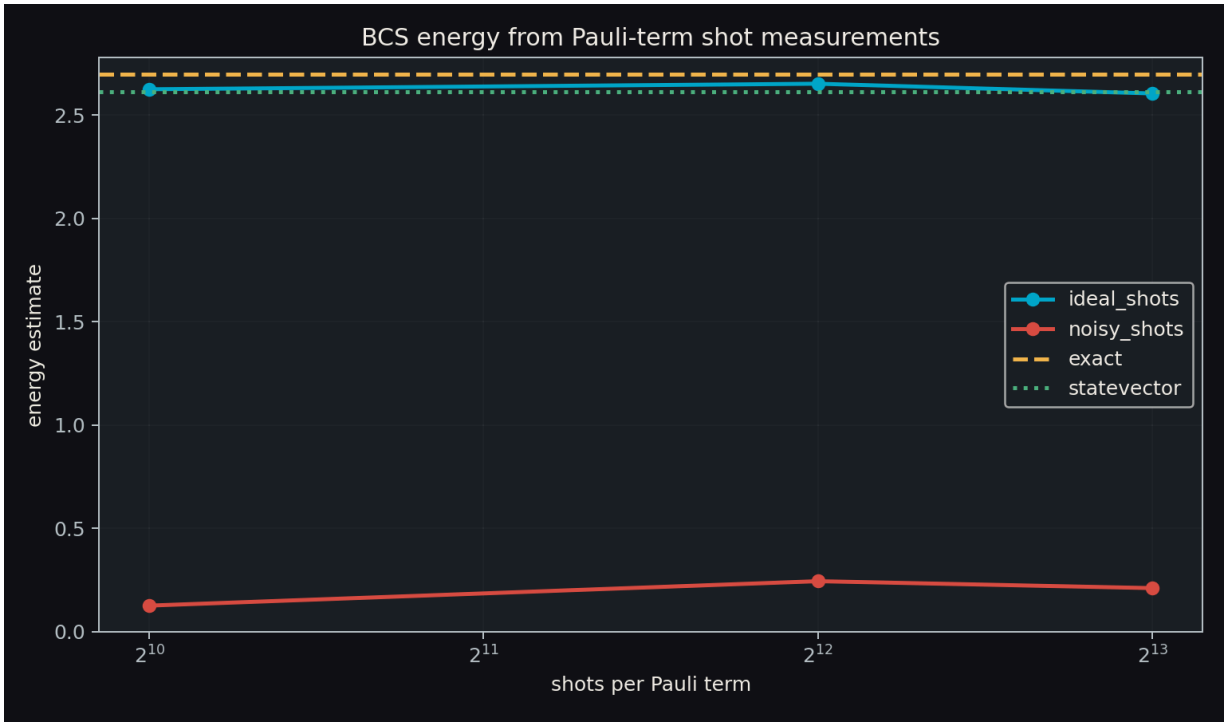
	mode	shots	exact_energy	statevector_energy	energy	abs_error_vs_statevec
0	ideal_shots	1024.0	2.7	2.61166	2.625391	0.013
1	noisy_shots	1024.0	2.7	2.61166	0.125879	2.485
2	ideal_shots	4096.0	2.7	2.61166	2.652319	0.040
3	noisy_shots	4096.0	2.7	2.61166	0.244165	2.367
4	ideal_shots	8192.0	2.7	2.61166	2.605103	0.006
5	noisy_shots	8192.0	2.7	2.61166	0.210583	2.401

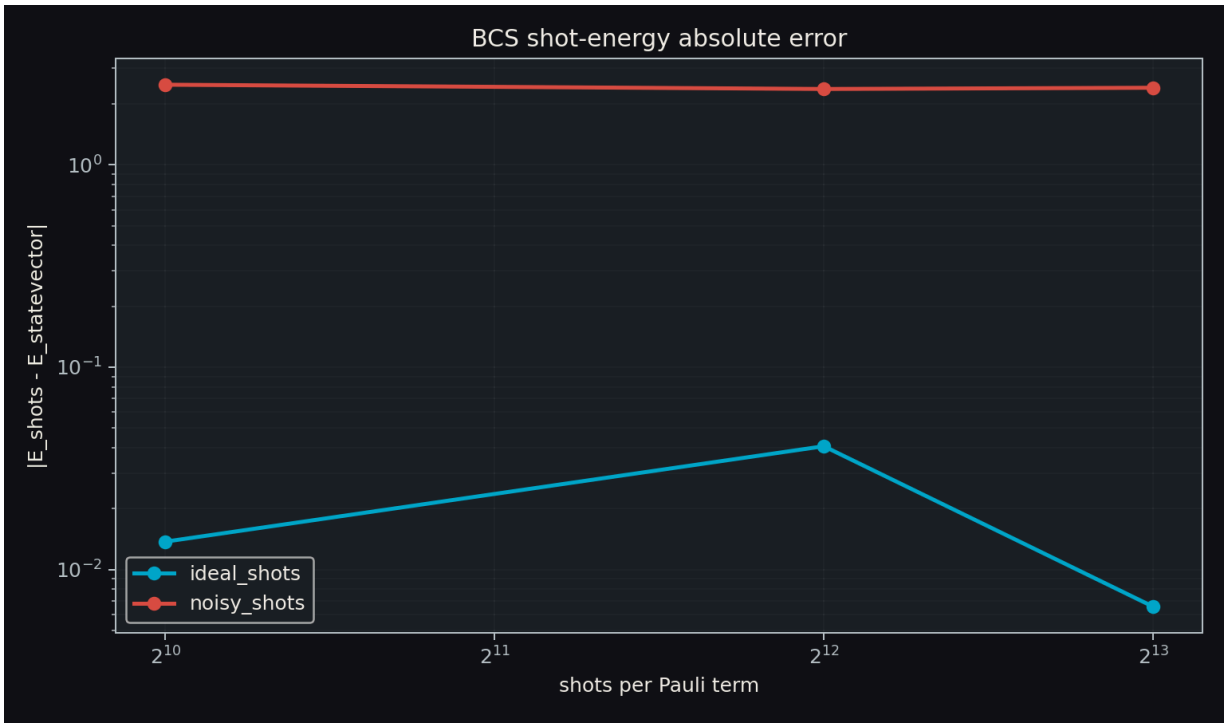
	mode	noise_scale	energy	abs_error_vs_statevector	noise_slope	fit_rmse
0	scaled_noise	1.0	1.425244	1.186416	NaN	NaN
1	scaled_noise	2.0	0.710791	1.900869	NaN	NaN
2	scaled_noise	3.0	0.418750	2.192910	NaN	NaN
3	linear_zne	0.0	1.858089	0.753571	-0.503247	0.099563

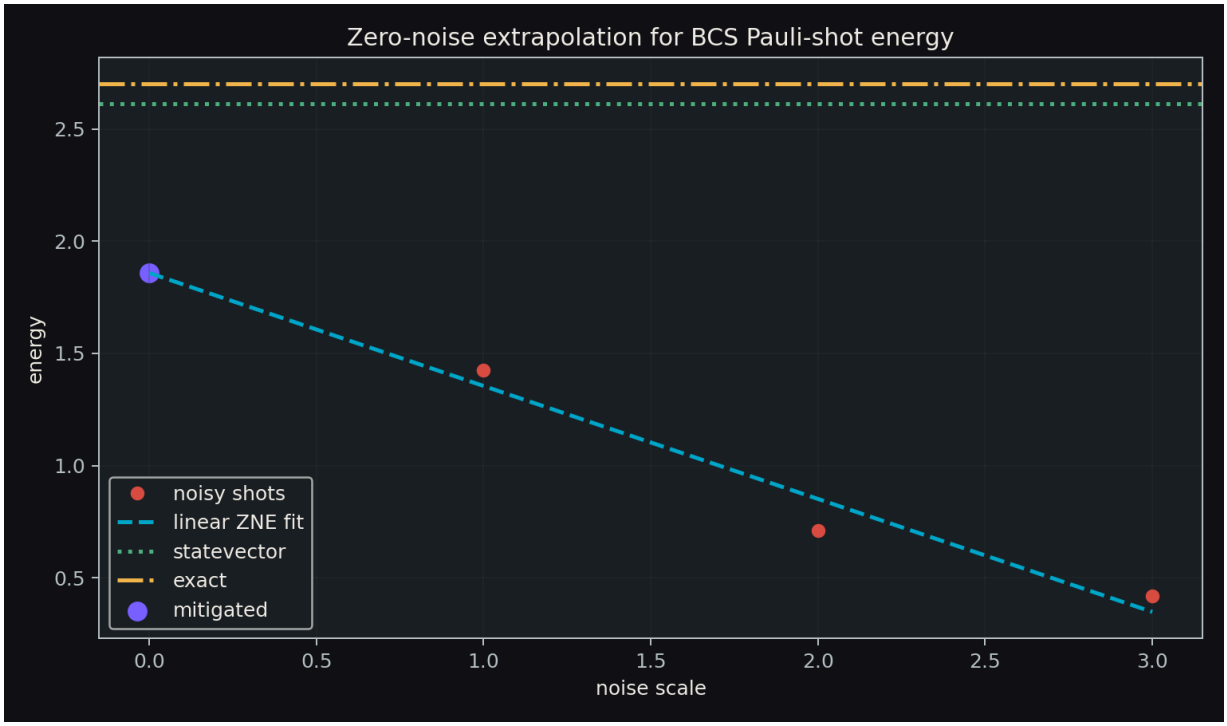
In [11]:
```
for fig in [
    'bcs_qiskit_shots_noise_total_pairs.png',
    'bcs_qiskit_shots_noise_occupations.png',
    'bcs_qiskit_energy_shots_values.png',
    'bcs_qiskit_energy_shots_error.png',
    'bcs_qiskit_error_mitigation_zne.png',
    'bcs_qiskit_error_mitigation_errors.png',
]:
    show_figure(fig, width=820)
```

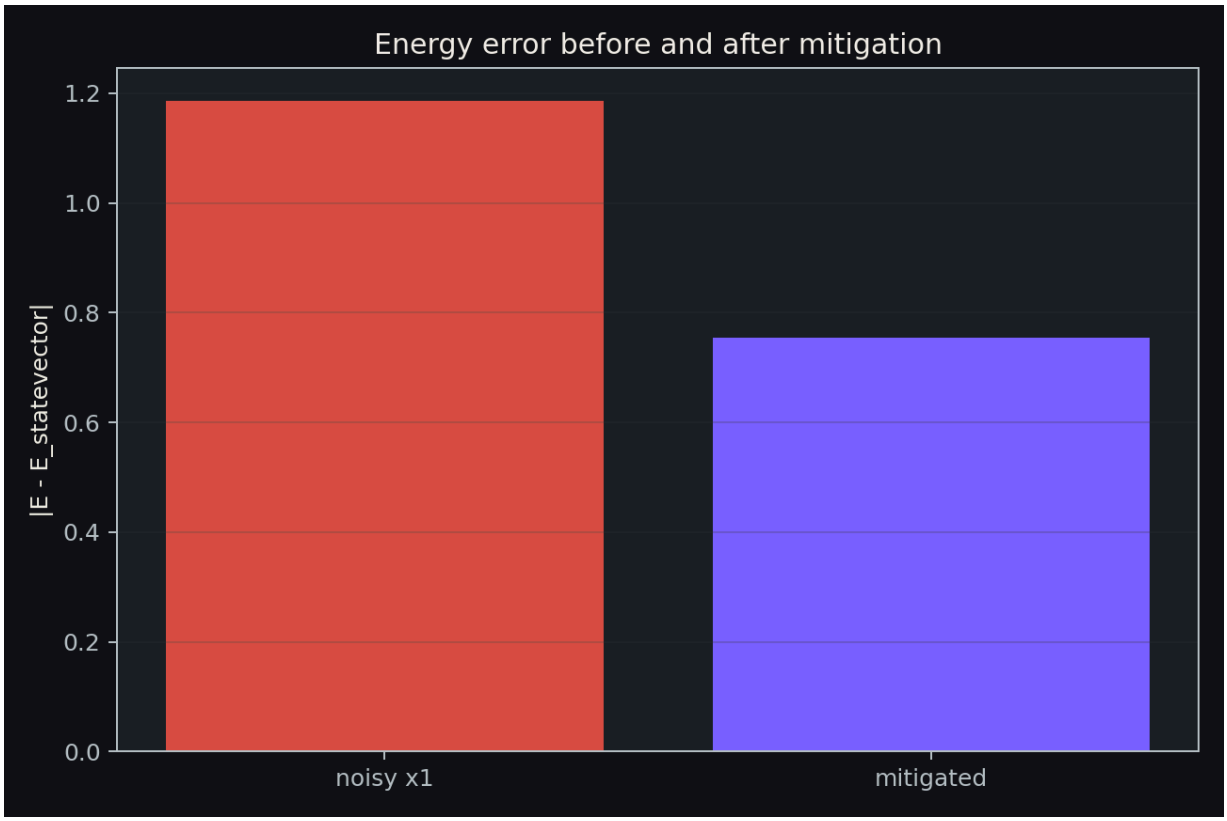

BCS Qiskit measurement: ideal vs noisy shots
estimated total pair number
2.00
1.99
1.98
1.97
1.96
1.95
2^{10}
2^{11}
2^{12}
2^{13}
shots
ideal_shots
noisy_shots
statevector


Measured BCS pair occupations
pair occupation
0.8
0.6
0.4
0.2
0.0
n_0
n_1
n_2
n_3
statevector
ideal_shots 8192
noisy_shots 8192


BCS energy from Pauli-term shot measurements
energy estimate
shots per Pauli term
ideal_shots
noisy_shots
exact
statevector
0.0
0.5
1.0
1.5
2.0
2.5
2^{10}
2^{11}
2^{12}
2^{13}


BCS shot-energy absolute error
|E_shots - E_statevector|
shots per Pauli term
ideal_shots
noisy_shots
10^{0}
10^{-1}
10^{-2}
2^{10}
2^{11}
2^{12}
2^{13}





Resumo numérico final

In [12]:
```
energy_8192 = energy[(energy['mode'] == 'ideal_shots') & (energy['shots'] ==
noisy_4096 = mitigation[(mitigation['mode'] == 'scaled_noise') & (mitigatior
zne = mitigation[mitigation['mode'] == 'linear_zne'].iloc[0]

paper_summary = pd.DataFrame([
    {'metric': 'final observable fidelity', 'value': final_obs['fidelity']},
```

```
    {'metric': 'final state error', 'value': final_obs['state_error']},
    {'metric': 'normal Green peak omega', 'value': summary_spectral.loc[0, '
    {'metric': 'anomalous Green peak omega', 'value': summary_spectral.loc[1
    {'metric': 'Nambu trace peak omega', 'value': summary_spectral.loc[2, 'c
    {'metric': '8192 ideal shot energy error vs statevector', 'value': energ
    {'metric': 'noisy energy error before mitigation', 'value': noisy_4096['
    {'metric': 'energy error after ZNE mitigation', 'value': zne['abs_error_
])
paper_summary
```

Out[12]:

	metric	value
0	final observable fidelity	0.999886
1	final state error	0.022627
2	normal Green peak omega	-1.044297
3	anomalous Green peak omega	3.132890
4	Nambu trace peak omega	-2.436692
5	8192 ideal shot energy error vs statevector	0.006558
6	noisy energy error before mitigation	1.186416
7	energy error after ZNE mitigation	0.753571

Regenerar resultados

Execute no terminal, a partir da raiz do repositório:

```
source .venv/bin/activate
python experiments/run_all.py
pandoc papers/bcs_quantum_simulation.md -s -o
papers/bcs_quantum_simulation.html
google-chrome --headless --disable-gpu --no-sandbox \
  --print-to-pdf="$PWD/papers/bcs_quantum_simulation.pdf" \
  "file://$PWD/papers/bcs_quantum_simulation.html"
```

Para validar o núcleo:

```
PYTHONPATH=src .venv/bin/python -m pytest -q tests
```